



**The CENTRE for EDUCATION
in MATHEMATICS and COMPUTING**

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**Topic Generator - Solution Set
Solutions**

1. The word **BANK** is painted exactly as shown on the outside of a clear glass window. Looking out through the window from the inside of the building, the word appears as
- (A) **BAИK** (B) **KNAB** (C) **BAИK** (D) **KNAB** (E) **KNAB**

Source: 2007 Gauss Grade 7 #9

Primary Topics: Geometry and Measurement

Secondary Topics: Transformations

Answer: D

Solution:

When the word **BANK** is viewed from the inside of the window, the letters appear in the reverse order and the letters themselves are all backwards, so the word appears as **KNAB**.

2. If the figure



is rotated 180° about point F , the result could be

(A)



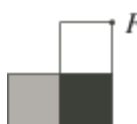
(B)



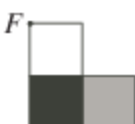
(C)



(D)



(E)



Source: 2009 Gauss Grade 7 #10

Primary Topics: Geometry and Measurement

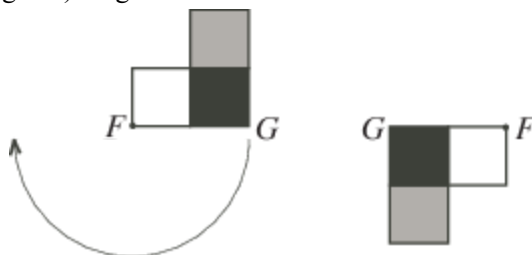
Secondary Topics: Transformations

Answer: C

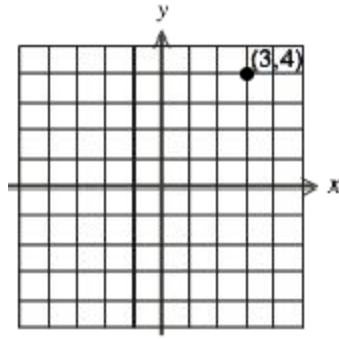
Solution:

Consider rotating the horizontal line segment FG (as shown below) 180° about point F .

A 180° rotation is half of a full rotation. Point F stays fixed, while segment FG rotates to the left of F (as does the rest of the diagram). Figure C shows the correct result.



3. If the point $(3, 4)$ is reflected in the x -axis, what are the coordinates of its image?



- (A) $(-4, 3)$ (B) $(-3, 4)$ (C) $(4, 3)$ (D) $(3, -4)$ (E) $(-3, -4)$

Source: 2011 Gauss Grade 7 #9

Primary Topics: Geometry and Measurement

Secondary Topics: Transformations

Answer: D

Solution:

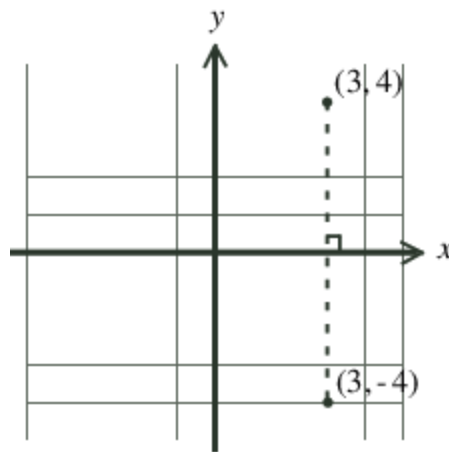
After reflecting the point $(3, 4)$ in the x -axis, the x -coordinate of the image will be the same as the x -coordinate of the original point, $x = 3$.

The original point is a distance of 4 from the x -axis.

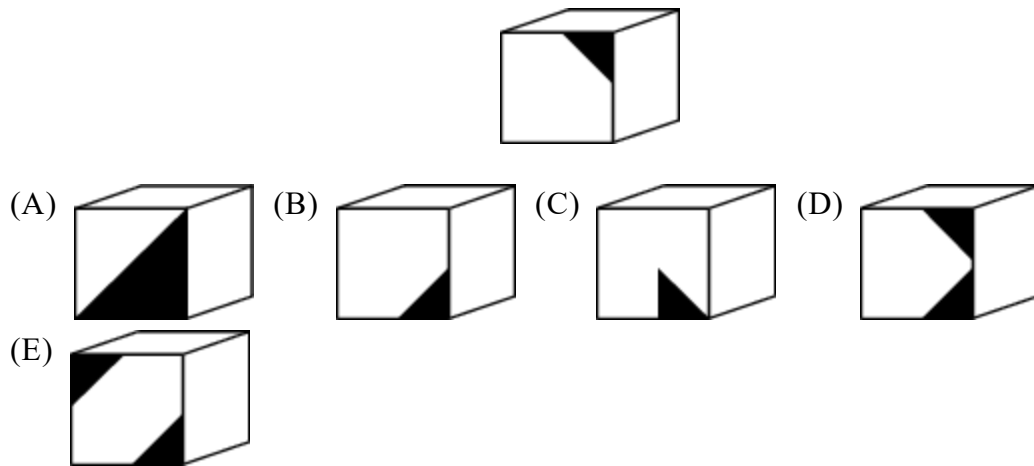
The image will be the same distance from the x -axis, but below the x -axis.

Thus, the image has y -coordinate -4 .

The coordinates of the image point are $(3, -4)$.



4. A cube has exactly one face painted as shown. The other five faces of the cube are not painted. If the cube is rolled, which of the following could be the same cube?



Source: 2016 Gauss Grade 7 #5

Primary Topics: Geometry and Measurement

Secondary Topics: Transformations

Answer: B

Solution:

Rolling the cube does not change the size of the painted triangle.

For this reason, we can eliminate answer (A).

Rolling the cube does not change the number of painted triangles.

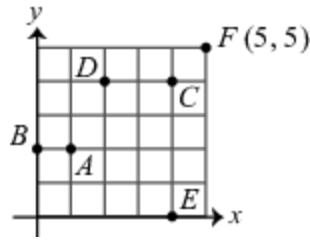
For this reason, we can eliminate answers (D) and (E).

Rolling the cube does not change the orientation of the painted triangle with respect to the face of the cube that it is painted on.

For this reason, we can eliminate answer (C).

Of the given answers, the cube shown in (B) is the only cube which could be the same as the cube that was rolled.

5. In the diagram, point F has coordinates $(5, 5)$. The point with coordinates $(2, 4)$ is located at



- (A) A (B) B (C) C (D) D (E) E

Source: 2019 Gauss Grade 7 #2

Primary Topics: Geometry and Measurement

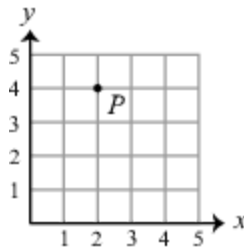
Secondary Topics: Coordinate Geometry

Answer: D

Solution:

Beginning at the origin $(0, 0)$, the point $(2, 4)$ is located right 2 units and up 4 units. The point $(2, 4)$ is located at D .

6. In the diagram shown, what are the coordinates of point P ?



- (A) $(4, 0)$ (B) $(2, 2)$ (C) $(2, 0)$ (D) $(4, 4)$ (E) $(2, 4)$

Source: 2020 Gauss Grade 7 #2

Primary Topics: Geometry and Measurement

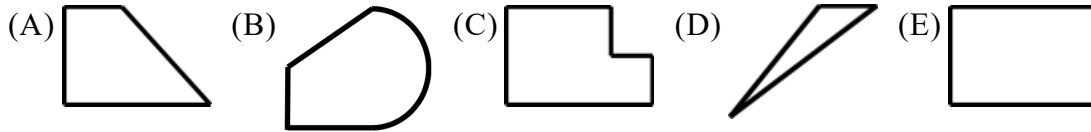
Secondary Topics: Coordinate Geometry

Answer: E

Solution:

Beginning at the origin $(0, 0)$, point P is located right 2 units (in the positive x direction) and up 4 units (in the positive y direction). Thus the coordinates of point P are $(2, 4)$.

7. Which of the following shapes has a vertical line of symmetry?



Source: 2024 Gauss Grade 7 #3

Primary Topics: Geometry and Measurement

Secondary Topics: Transformations

Answer: E

Solution:

If a figure has a vertical line of symmetry, then when it is reflected in (or folded along) this vertical line, the two halves of the figure are identical to one another. Of the given shapes, such a vertical line is only possible for (E).

8. The words "P U G F O R S A L E" are written on a store window. How many of these ten letters look the same when viewed from both sides of the window?

- (A) 3 (B) 4 (C) 5 (D) 6 (E) 7

Source: 2025 Gauss Grade 7 #10

Primary Topics: Geometry and Measurement

Secondary Topics: Transformations

Answer: A

Solution:

When viewed from the opposite side of the window, text appears reflected horizontally.

That is, P U G F O R S A L E appears as . The letters that look the same when viewed from both sides of the window are A, O and U and so there are 3 such letters.

9. The letter P is written in a 2×2 grid of squares as shown:  A combination of rotations about the centre of the grid and reflections in the two lines through the centre achieves the result: 

When the same combination of rotations and reflect is applied to , the result is

- (A)  (B)  (C)  (D)  (E) 

Source: 2006 Gauss Grade 7 #20

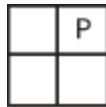
Primary Topics: Geometry and Measurement

Secondary Topics: Transformations

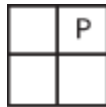
Answer: B

Solution:

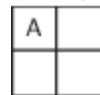
One possible way to transform the initial position of the P to the final position of the P is to reflect the grid in the vertical line in the middle to obtain



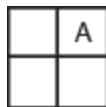
and then rotate the grid 90° counterclockwise about the centre to obtain



Applying these transformations to the grid containing the A, we obtain



and then



(There are many other possible combinations of transformations which will produce the same resulting image with the P; each of these combinations will produce the same result with the A.)

10. Using two transformations, the letter **R** is changed as shown: **R** → **Я** → **Ɔ**.
 Using the same two transformations, the letter **L** is changed as shown: **L** → **┐** → **┌**.
 Using the same two transformations, the letter **G** is changed to
 (A) **G** (B) **Ɔ** (C) **Я** (D) **Ɔ** (E) **Г**

Source: 2008 Gauss Grade 7 #15

Primary Topics: Geometry and Measurement

Secondary Topics: Transformations

Answer: D

Solution:

The first transformation is a 180° rotation of the letter, which gives **G** → **Ɔ**.

The second transformation is a reflection across a vertical axis, which gives **G** → **Ɔ**.

11. A 4×4 square grid can be entirely covered by three non-overlapping pieces made from 1×1 squares. If the first two pieces are  and , the third piece is
 (A)  (B)  (C)  (D)  (E) 

Source: 2008 Gauss Grade 7 #19

Primary Topics: Geometry and Measurement

Secondary Topics: Transformations

Answer: C

Solution:

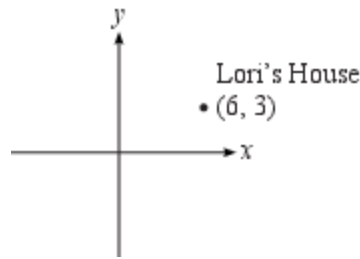
After some experimentation, the only way in which the two given pieces can be put together to stay within a 4×4 grid and so that one of the given choices can fit together with them is to rotate the second piece by 90° clockwise, and combine to obtain



Therefore, the missing piece is



12. The diagram shows Lori's house located at $(6, 3)$. If Alex's house is located at $(-2, -4)$, what translation is needed to get from Lori's house to Alex's house?



- (A) 4 units left, 1 unit up (B) 8 units right, 7 units up (C) 4 units left, 1 unit down
(D) 8 units left, 7 units down (E) 7 units right, 8 units down

Source: 2014 Gauss Grade 7 #15

Primary Topics: Geometry and Measurement

Secondary Topics: Transformations | Coordinate Geometry

Answer: D

Solution:

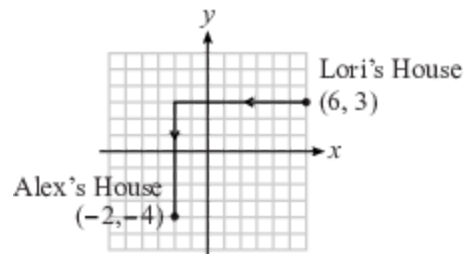
The horizontal translation needed to get from Lori's house to Alex's house is the difference between the x -coordinate of Lori's house, 6, and the x -coordinate of Alex's house, -2 , or

$$6 - (-2) = 6 + 2 = 8.$$

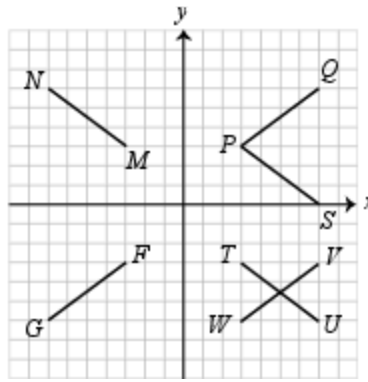
The vertical translation needed to get from Lori's house to Alex's house is the difference between the y -coordinate of Lori's house, 3, and the y -coordinate of Alex's house, -4 , or $3 - (-4) = 3 + 4 = 7$.

From Lori's house, Alex's house is left and down.

Therefore the translation needed to get from Lori's house to Alex's house is 8 units left and 7 units down.



13. In the graph shown, which of the following represents the image of the line segment PQ after a reflection across the x -axis?



- (A) PS (B) TU (C) MN (D) WV (E) FG

Source: 2015 Gauss Grade 7 #18

Primary Topics: Geometry and Measurement

Secondary Topics: Transformations

Answer: B

Solution:

To find the image of PQ , we reflect points P and Q across the x -axis, then join them.

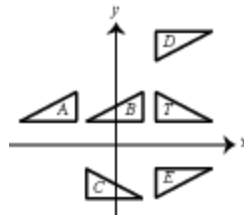
Since P is 3 units above the x -axis, then the reflection of P across the x -axis is 3 units below the x -axis at the same x -coordinate.

That is, point T is the image of P after it is reflected across the x -axis.

Similarly, after a reflection across the x -axis, the image of point Q will be 6 units below the x -axis but have the same x -coordinate as Q .

That is, point U is the image of Q after it is reflected across the x -axis. Therefore, the line segment TU is the image of PQ after it is reflected across the x -axis.

14. Triangle T is reflected once. Which of the following triangles *cannot* be this reflection of triangle T ?



- (A) A (B) B (C) C (D) D (E) E

Source: 2016 Gauss Grade 7 #17

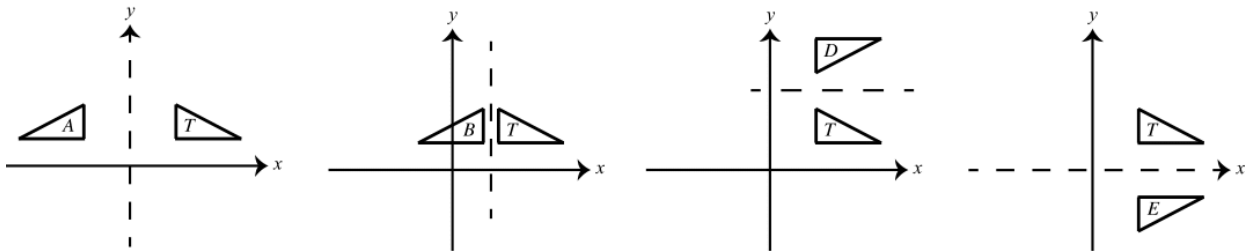
Primary Topics: Geometry and Measurement

Secondary Topics: Transformations

Answer: C

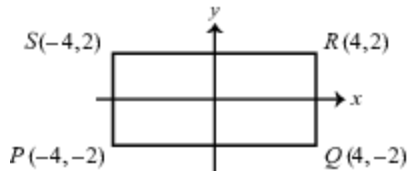
Solution:

Each of the following four diagrams shows the image of triangle T after its reflection in the dotted line.



Thus, each of the triangles labelled A , B , D , and E is a single reflection of triangle T in some line. The triangle labelled C is the only triangle that cannot be a reflection of triangle T .

15. In the diagram, what is the area of rectangle $PQRS$?



- (A) 36 (B) 32 (C) 40 (D) 20 (E) 44

Source: 2019 Gauss Grade 7 #12

Primary Topics: Geometry and Measurement

Secondary Topics: Area | Coordinate Geometry

Answer: B

Solution:

The width of rectangle $PQRS$ is the horizontal distance between points P and Q (or points S and R), since these points have equal y -coordinates.

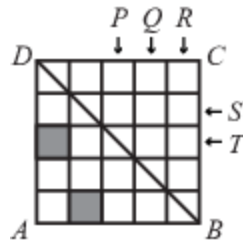
This distance is equal to the difference between their x -coordinates or $4 - (-4) = 8$.

Similarly, the height of rectangle $PQRS$ is equal the vertical distance between points S and P (or points R and Q), since these points have equal x -coordinates.

This distance is equal to the difference between their y -coordinates or $2 - (-2) = 4$.

The area of rectangle $PQRS$ is $8 \times 4 = 32$.

16. In the diagram, which of the following squares should be shaded to make BD a line of symmetry of square $ABCD$?



- (A) P and S (B) Q and S (C) P and T (D) Q and T (E) P and R

Source: 2020 Gauss Grade 7 #16

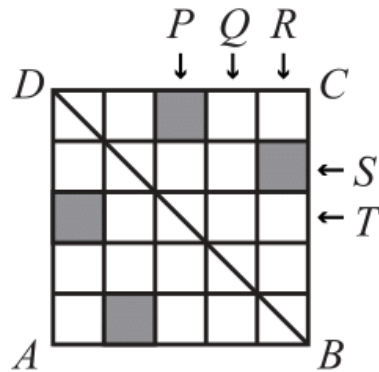
Primary Topics: Geometry and Measurement

Secondary Topics: Coordinate Geometry

Answer: A

Solution:

For BD to be a line of symmetry of $ABCD$, the squares labelled P and S should be shaded.



To see why this is, consider that the reflection in BD of vertex A is vertex C , and so the reflection in BD of side DA is side DC .

Further, the reflection in BD of the first column of square $ABCD$ is the first row of square $ABCD$.

Thus, the reflection in BD of the shaded square in row 3, column 1 is the square in row 1, column 3 (the square labelled P).

More generally, the reflection in BD of the square in row r , column c is the square in row c , column r . Thus, the reflection in BD of the shaded square in row 5, column 2 is the square in row 2, column 5 (the square labelled S).

17. The letter **A** has a vertical line of symmetry and the letter **B** does not. How many of the letters **H, L, O, R, X, D, P, E** have a vertical line of symmetry?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Source: 2021 Gauss Grade 7 #12

Primary Topics: Geometry and Measurement

Secondary Topics: Transformations

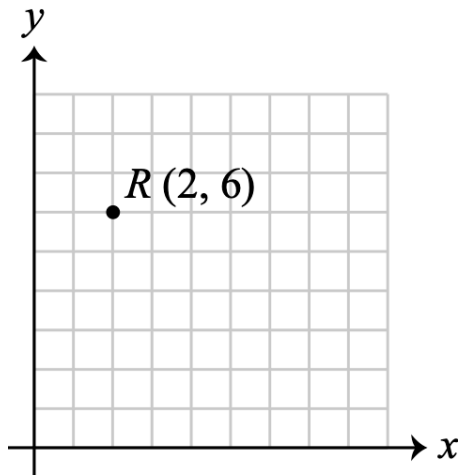
Answer: C

Solution:

If a letter is folded along its vertical line of symmetry, both halves of the letter would match exactly. Three of the given letters, **H, O** and **X**, have a vertical line of symmetry.

H L O R X D P E

18. The coordinates of R are $(2, 6)$, as shown. After which of these translations will R move to the point $(7, 0)$?



- (A) right (9), down (6) (B) left (5), up (6) (C) right (6), down (5)
(D) left (6), down (5) (E) right (5), down (6)

Source: 2025 Gauss Grade 7 #11

Primary Topics: Geometry and Measurement

Secondary Topics: Coordinate Geometry | Transformations

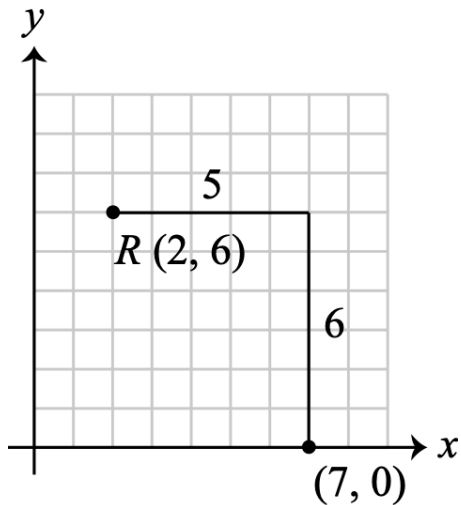
Answer: E

Solution:

The x -coordinate of $(2, 6)$ is 5 less than the x -coordinate of $(7, 0)$, and so R must be translated right 5

. The y -coordinate of $(2, 6)$ is 6 more than the y -coordinate of $(7, 0)$, and so R must be translated down 6.

Thus, the translation right 5, down 6 will move $R(2, 6)$ to $(7, 0)$.



19. A dot starts at $(20, 19)$. It can move one unit vertically or horizontally to one of the points $(21, 19)$, $(19, 19)$, $(20, 20)$, or $(20, 18)$. From there it can move two units in either direction that is perpendicular to the first move. All moves thereafter increase in length by one unit (three units, four units, five units, etc.) and must be perpendicular to the direction of the previous move. The dot stops after ten moves. Which of the following final locations is *not* possible?
- (A) $(27, 33)$ (B) $(30, 40)$ (C) $(21, 21)$ (D) $(42, 44)$ (E) $(37, 37)$

Source: 2019 Gauss Grade 7 #24

Primary Topics: Algebra and Equations | Geometry and Measurement | Number Sense

Secondary Topics: Patterning/Sequences/Series | Coordinate Geometry

Answer: B

Solution:

Solution 1

The ten moves have lengths 1, 2, 3, 4, 5, 6, 7, 8, 9, 10.

If the first move is vertical, then the five vertical moves have lengths 1, 3, 5, 7, 9 and the five horizontal moves have lengths 2, 4, 6, 8, 10.

If the first move is horizontal, then the five horizontal moves have lengths 1, 3, 5, 7, 9 and the five vertical moves have lengths 2, 4, 6, 8, 10.

If a horizontal move is to the right, then the length of the move is added to the x -coordinate.

If a horizontal move is to the left, then the length of the move is subtracted from the x -coordinate.

If a vertical move is up, then the length of the move is added to the y -coordinate.

If a vertical move is down, then the length of the move is subtracted from the y -coordinate.

Therefore, once the ten moves have been made, the change in one of the coordinates is a combination of adding and subtracting 1, 3, 5, 7, 9 and the change in the other coordinate is a combination of adding and subtracting 2, 4, 6, 8, 10.

For example, if the dot moves right 1, down 2, right 3, up 4, right 5, down 6, right 7, up 8, left 9, and up 10, then its final x -coordinate is

$$20 + 1 + 3 + 5 + 7 - 9 = 27$$

and its final y -coordinate is

$$19 - 2 + 4 - 6 + 8 + 10 = 33$$

making its final location (27, 33), which is choice (A). This means that choice (A) is not the answer.

We note that, in the direction with the moves of even length, the final change in coordinate will be even, since whenever we add and subtract even integers, we obtain an even integer.

In the other direction, the final change in coordinate will be odd, since adding or subtracting an odd number of odd integers results in an odd integer. (Since odd plus odd is even and odd minus odd is even, then after two moves of odd length, the change to date is even, and after four moves of odd length, the change to date is still even, which means that the final change after the fifth move of odd length is completed must be odd, since even plus or minus odd is odd.)

In the table below, these observations allow us to determine in which direction to put the moves of odd length and in which direction to put the moves of even length.

Choice	Change in x	Change in y	Horizontal moves	Vertical moves
(A) (27, 33)	7	14	$1 + 3 + 5 + 7 - 9 = 7$	$-2 + 4 - 6 + 8 + 10 = 14$
(B) (30, 40)	10	21	$2 - 4 - 6 + 8 + 10 = 10$	
(C) (21, 21)	1	2	$1 - 3 + 5 + 7 - 9 = 1$	$2 + 4 - 6 - 8 + 10 = 2$
(D) (42, 44)	22	25	$2 - 4 + 6 + 8 + 10 = 22$	$1 + 3 + 5 + 7 + 9 = 25$
(E) (37, 37)	17	18	$-1 - 3 + 5 + 7 + 9 = 17$	$2 + 4 - 6 + 8 + 10 = 18$

Since each of the locations (A), (C), (D), and (E) is possible, then the location that is not possible must be (B).

We note that in the case of (B), it is the change in the y -coordinate that is not possible to make.

In other words, we cannot obtain a total of 21 by adding and subtracting 1, 3, 5, 7, 9.

Can you see why?

Solution 2

Let a be the horizontal change from the initial to the final position of the point and let b be the vertical change from the initial to the final position.

For example, if $a = -5$ and $b = 6$, the point's final position is 5 units to the left and 6 units up from its original position of (20, 19).

Notice that if (x, y) is the final position of the point, then a and b can be calculated as $a + 20 = x$ or $a = x - 20$ and $b + 19 = y$ or $b = y - 19$.

First, assume the initial move is in the horizontal direction.

This means the second move will be in the vertical direction, the third will be horizontal, and so on.

In all, the first, third, fifth, seventh, and ninth moves will be horizontal and the others will be vertical.

Also, the first move is by one unit, the second is by two units, the third is by three units, and so on, so the horizontal moves are by 1, 3, 5, 7, and 9 units.

Each of these moves is either to the left or the right.

If all of the horizontal moves are to the right, then $a = 1 + 3 + 5 + 7 + 9 = 25$.

If the point moves, left on the first move, right on the third, then left on the fifth, seventh, and ninth moves, a will be $-1 + 3 - 5 - 7 - 9 = -19$.

In this case, the final position is 19 units to the left of the initial position (and has potentially moved up or down, as well).

In each of the five horizontal moves, the point either moves to the left or to the right.

This means there are two choices (left or right) for each of the five horizontal moves, so there are $2 \times 2 \times 2 \times 2 \times 2 = 32$ possible configurations similar to the examples above.

Going through these carefully, the table below computes all possible values of a :

$$\begin{aligned}
 1 + 3 + 5 + 7 + 9 &= 25 \\
 -1 + 3 + 5 + 7 + 9 &= 23 \\
 1 - 3 + 5 + 7 + 9 &= 19 \\
 -1 - 3 + 5 + 7 + 9 &= 17 \\
 1 + 3 - 5 + 7 + 9 &= 15 \\
 -1 + 3 - 5 + 7 + 9 &= 13 \\
 1 - 3 - 5 + 7 + 9 &= 9 \\
 -1 - 3 - 5 + 7 + 9 &= 7 \\
 1 + 3 + 5 - 7 + 9 &= 11 \\
 -1 + 3 + 5 - 7 + 9 &= 9 \\
 1 - 3 + 5 - 7 + 9 &= 5 \\
 -1 - 3 + 5 - 7 + 9 &= 3 \\
 1 + 3 - 5 - 7 + 9 &= 1 \\
 -1 + 3 - 5 - 7 + 9 &= -1 \\
 1 - 3 - 5 - 7 + 9 &= -5 \\
 -1 - 3 - 5 - 7 + 9 &= -7 \\
 1 + 3 + 5 + 7 - 9 &= 7 \\
 -1 + 3 + 5 + 7 - 9 &= 5 \\
 1 - 3 + 5 + 7 - 9 &= 1 \\
 -1 - 3 + 5 + 7 - 9 &= -1 \\
 1 + 3 - 5 + 7 - 9 &= -3 \\
 -1 + 3 - 5 + 7 - 9 &= -5 \\
 1 - 3 - 5 + 7 - 9 &= -9 \\
 -1 - 3 - 5 + 7 - 9 &= -11 \\
 1 + 3 + 5 - 7 - 9 &= -8 \\
 -1 + 3 + 5 - 7 - 9 &= -9 \\
 1 - 3 + 5 - 7 - 9 &= -13 \\
 -1 - 3 + 5 - 7 - 9 &= -15 \\
 1 + 3 - 5 - 7 - 9 &= -17 \\
 -1 + 3 - 5 - 7 - 9 &= -19 \\
 1 - 3 - 5 - 7 - 9 &= -23 \\
 -1 - 3 - 5 - 7 - 9 &= -25
 \end{aligned}$$

Some numbers appear more than once, but after inspecting the list, we see that when the first move is in the horizontal direction, a can be any odd number between -25 and 25 inclusive except for -21 and 21 .

When the first move is horizontal, the second, fourth, sixth, eighth, and tenth moves will be vertical and have lengths 2, 4, 6, 8, and 10 units.

Using the same idea as the previous case, all 32 possible values of b when the first move is horizontal are calculated as follows:

$$\begin{aligned}
&2 + 4 + 6 + 8 + 10 = 30 \\
&-2 + 4 + 6 + 8 + 10 = 26 \\
&2 - 4 + 6 + 8 + 10 = 22 \\
&-2 - 4 + 6 + 8 + 10 = 18 \\
&2 + 4 - 6 + 8 + 10 = 18 \\
&-2 + 4 - 6 + 8 + 10 = 14 \\
&2 - 4 - 6 + 8 + 10 = 10 \\
&-2 - 4 - 6 + 8 + 10 = 6 \\
&2 + 4 + 6 - 8 + 10 = 14 \\
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&2 - 4 + 6 - 8 + 10 = 6 \\
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&2 + 4 - 6 - 8 + 10 = 2 \\
&-2 + 4 - 6 - 8 + 10 = -2 \\
&2 - 4 - 6 - 8 + 10 = -6 \\
&-2 - 4 - 6 - 8 + 10 = -10 \\
&2 + 4 + 6 + 8 - 10 = 10 \\
&-2 + 4 + 6 + 8 - 10 = 6 \\
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&-2 + 4 + 6 - 8 - 10 = -10 \\
&2 - 4 + 6 - 8 - 10 = -14 \\
&-2 - 4 + 6 - 8 - 10 = -18 \\
&2 + 4 - 6 - 8 - 10 = -18 \\
&-2 + 4 - 6 - 8 - 10 = -22 \\
&2 - 4 - 6 - 8 - 10 = -26 \\
&-2 - 4 - 6 - 8 - 10 = -30
\end{aligned}$$

Again, some numbers appear more than once, but examination of the table shows that when the first move is horizontal, the possible values of b are the even numbers between -30 and 30 inclusive which are not multiples of 4 .

We now begin to inspect the possible answers. The point in (A) is $(27, 33)$.

In this case, we have that $a = 27 - 20 = 7$ and $b = 33 - 19 = 14$.

These values for a and b fit the descriptions above, so $(27, 33)$ is a possible final position.

The point $(21, 21)$ in (C) has $a = 1$ and $b = 2$, which means $(21, 21)$ is also a possible final position for the point.

The point $(37, 37)$ in (E) has $a = 17$ and $b = 18$, so this point is also a possible final position.

We have shown that the answer must be either (B) or (D).

Note that in both cases we have that a is even and b is odd, which means that if either of these points is a possible final position, the first move cannot have been horizontal, which means it must have been vertical.

If the first move is vertical, then the third, fifth, seventh, and ninth moves are also vertical, and the other moves are horizontal.

Going through similar analysis as before, we will see that the restrictions on a and b have switched. That is, a must be an even number between -30 and 30 inclusive which is not a multiple of 4 , and b must be an odd number between -25 and 25 inclusive other than -21 and 21 .

The point $(42, 44)$ in (D) has $a = 22$ and $b = 25$, which is possible with a vertical first move.

However, the point $(30, 40)$ in (B) has $a = 10$ and $b = 21$.

Since 21 is not a possible value for b , then $(30, 40)$ is not a possible final position.

